Real-Data Study of Weekend vs. Weekday Shopping Spending

Short Summary

Is there a significant difference between weekend and weekday shopper spending? We show this question is a special case of simple linear regression on a single binary indicator variable, with the regression t-statistic proven to coincide exactly with the classical pooled two-sample t-statistic under variance homogeneity (Theorem 3.1). Relaxing that assumption, the Behrens–Fisher problem and Welch's approximate solution are derived in full, including the exact algebraic coincidence between the pooled and Welch t-statistics in balanced designs, and the precise sense in which their p-values still differ (Remark 4.1). These results are applied to a real dataset of 541,909 timestamped UK retail transactions (Section 5): comparing a realistic $n_1 = n_2 = 500$ sample against the full population it was drawn from shows that the sample can lead us to the wrong conclusion: its p-value ($p \approx 0.31$) may lead us to conclude that weekend and weekday shoppers spend similarly, when the population-level p-value ($p < 0.05$) shows a real, small difference actually exists. Furthermore, the divergence between the pooled and Welch tests grows with both sample-size imbalance and the genuine possibility of variance heterogeneity in the data.

Contents

1 Introduction

1.1 The Motivating Problem

We are interested in shopping habits: specifically, whether people spend differently on weekends than on weekdays. If a social researcher were to investigate it by taking finite samples of shoppers and measuring what they spend, could they assert that a genuine difference in shopping habits exists between the two groups, and how would that assertion be quantified rather than just guessed at? Concretely: suppose $n_1 = 500$ weekend shoppers are sampled and found to spend \$368.86 on average with standard deviation \$382.24, while $n_2 = 500$ weekday shoppers are sampled and found to spend \$458.64 on average with standard deviation \$686.09. Is there sufficient evidence to claim a significant difference in average spending between the two groups, and what is the attained significance level?

We look at two samples drawn from different populations, and we want to know how different these populations actually are, based only on a sample analysis of each. Along the analysis we show that this problem is not really a separate technique from linear regression: it *is* a linear regression, in disguise, with a single 0/1 predictor. Seeing the two frameworks as one and the same clarifies exactly what the assumption of "variance homogeneity" gives us, and what breaks when it fails: we then need a different test, such as Welch's test, to properly account for the unequal variances.

2 The Two-Sample Model

Consider two populations with unknown means μ_1, μ_2 and unknown variances σ_1^2, σ_2^2; neither population is observed directly, and inference proceeds from a finite random sample of each. Specializing to the problem at hand, take the two populations to be the amount spent by a weekend shopper and the amount spent by a weekday shopper, with *each shopper* modeled as an independent, identically distributed normal draw from the relevant population:

$$\begin{aligned} X_1, \ldots, X_{n_1} &\overset{iid}{\sim} N(\mu_1, \sigma_1^2) \quad \text{(weekend spending)}, \\ Y_1, \ldots, Y_{n_2} &\overset{iid}{\sim} N(\mu_2, \sigma_2^2) \quad \text{(weekday spending)}, \end{aligned} \tag{1}$$

with the two samples mutually independent.

Proposition 2.1 (Distribution of $\bar{X} - \bar{Y}$, arbitrary variances)
Define

$$\bar{X} = \frac{1}{n_1} \sum_{i=1}^{n_1} X_i, \qquad \bar{Y} = \frac{1}{n_2} \sum_{j=1}^{n_2} Y_j$$

for the sample means of the two groups. We then have:

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right). \tag{2}$$

Proof.

Since $X_1, \ldots, X_{n_1}$ are i.i.d. $N(\mu_1, \sigma_1^2)$, a linear combination of independent normals is itself normal, with mean and variance given by the same linear combination of the individual means and (for variance) squared coefficients:

$$\bar{X} = \sum_{i=1}^{n_1} \frac{1}{n_1} X_i \quad \Longrightarrow \quad \mathrm{E}[\bar{X}] = \sum_{i=1}^{n_1} \frac{1}{n_1} \mu_1 = \mu_1, \qquad \mathrm{Var}(\bar{X}) = \sum_{i=1}^{n_1} \frac{1}{n_1^2} \sigma_1^2 = \frac{\sigma_1^2}{n_1},$$

this gives

$$\bar{X} \sim N(\mu_1, \sigma_1^2/n_1), \qquad \bar{Y} \sim N(\mu_2, \sigma_2^2/n_2)$$

Since the two samples are independent, $\bar{X}$ and $\bar{Y}$ are independent. A difference of independent normal random variables is again normal, with mean and variance given by:

$$\mathrm{E}[\bar{X} - \bar{Y}] = \mu_1 - \mu_2, \qquad \mathrm{Var}(\bar{X} - \bar{Y}) = \mathrm{Var}(\bar{X}) + \mathrm{Var}(\bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

This gives $\bar{X} - \bar{Y} \sim N\big(\mu_1 - \mu_2,\ \sigma_1^2/n_1 + \sigma_2^2/n_2\big)$. □

Standardizing (2) gives

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0, 1). \tag{3}$$

Testing $H_0 : \mu_1 = \mu_2$ with Z is not possible in practice: Z depends on σ_1^2, σ_2^2, and these are unknown. The solution is to replace σ_1^2, σ_2^2 with the sample variances, given by: [1]

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \qquad S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2;$$

Definition 2.1 (Variance homogeneity)
The samples in (1) satisfy *variance homogeneity* if

$$\sigma_1^2 = \sigma_2^2 = \sigma^2 \tag{4}$$

for some common, unknown σ^2.

Theorem 2.1 (Pooled two-sample t-statistic)
Under the two-sample setup (1) with variance homogeneity (4), for $H_0 : \mu_1 = \mu_2$ against $H_a : \mu_1 \neq \mu_2$,

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2} \quad \text{under } H_0, \qquad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}. \tag{5}$$

Proof.

Specializing Proposition 2.1 to $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2,\ \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right).$$

Standardizing this normal random variable gives

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}} \sim N(0, 1).$$

With a single common σ^2, the two group variances estimate the *same* quantity, so it is legitimate to combine them into one pooled estimate. From normal-sample theory, each group's rescaled sample variance is individually chi-square distributed (see Appendix A):

$$\frac{(n_1 - 1)S_1^2}{\sigma^2} \sim \chi^2_{n_1-1}, \qquad \frac{(n_2 - 1)S_2^2}{\sigma^2} \sim \chi^2_{n_2-1},$$

[1] These are unbiased estimators of the population variances: $\mathrm{E}[S_1^2] = \sigma_1^2$ and $\mathrm{E}[S_2^2] = \sigma_2^2$. The $n_i - 1$ divisor (rather than n_i) is exactly what makes this true: dividing by n_i instead would systematically underestimate σ_i^2, since $\bar{X}$ (or $\bar{Y}$) is itself estimated from the same data, using up one degree of freedom.

independently of one another. A sum of independent chi-square random variables is again chi-square, with degrees of freedom adding:

$$W := \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \frac{(n_1+n_2-2)S_p^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}.$$

Without a common σ^2 to divide through by, $(n_1-1)S_1^2/\sigma_1^2$ and $(n_2-1)S_2^2/\sigma_2^2$ are each individually chi-square, but their sum is not proportional to any chi-square distribution once $\sigma_1^2 \neq \sigma_2^2$.

For normal samples, the sample mean and sample variance are independent, so Z and W are independent (see Appendix A). By the definition of the t-distribution, if $Z \sim N(0,1)$ and $W \sim \chi^2_k$ are independent, then (see Appendix B):

$$T = \frac{Z}{\sqrt{W/k}} \sim t_k.$$

Substituting Z and W above, with $k = n_1 + n_2 - 2$,

$$T = \frac{\dfrac{(\bar{X}-\bar{Y}) - (\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+1/n_2}}}{\sqrt{S_p^2/\sigma^2}}.$$

The two factors of σ cancel exactly, leaving

$$T = \frac{(\bar{X}-\bar{Y}) - (\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \sim t_{n_1+n_2-2}.$$

Under $H_0 : \mu_1 = \mu_2$, the numerator's shift term vanishes, and this is exactly (5). Under the null hypothesis H_0, $T_{n_1+n_2-2}$ is the random variable whose distribution is the single observed value computed from the actual data. The p-value quantifies, under the assumption that H_0 holds, the probability of observing a test statistic at least as extreme as the one actually obtained. For the two-sided alternative hypothesis $H_a : \mu_1 \neq \mu_2$, the relevant event is $|T_{n_1+n_2-2}| > |t_{\text{obs}}|$:

$$p = P(|T_{n_1+n_2-2}| > |t_{\text{obs}}|)\,.$$

This event splits into two disjoint tails, $T_{n_1+n_2-2} > |t_{\text{obs}}|$ or $T_{n_1+n_2-2} < -|t_{\text{obs}}|$, so

$$p = P(T_{n_1+n_2-2} > |t_{\text{obs}}|) + P(T_{n_1+n_2-2} < -|t_{\text{obs}}|)\,.$$

Since the t-distribution is symmetric about 0, both tails have equal probability, $P(T_{n_1+n_2-2} < -|t_{\text{obs}}|) = P(T_{n_1+n_2-2} > |t_{\text{obs}}|)$, so

$$p = 2P(T_{n_1+n_2-2} > |t_{\text{obs}}|)\,. \qquad \square$$

3 Reformulating as Linear Regression

3.1 The Dummy-Variable Model

Stack all $n = n_1 + n_2$ observations into a single response vector, and define a group indicator

$$D_i = \begin{cases} 1 & \text{if observation } i \text{ is a weekend shopper} \\ 0 & \text{if observation } i \text{ is a weekday shopper.} \end{cases}$$

Consider the simple linear regression

$$Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i, \qquad i = 1, \ldots, n. \tag{6}$$

In matrix form, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with

$$\mathbf{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & D_1 \\ \vdots & \vdots \\ 1 & D_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The ordinary least squares (OLS) estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}, \tag{7}$$

the general derivation of which is given in Appendix C. This is an ordinary regression with one intercept and one 0/1 predictor.

Theorem 3.1 (Regression = two-sample comparison)
Let $\hat{\beta}_0, \hat{\beta}_1$ be the OLS estimators (7) of model (6). Then

$$\hat{\beta}_0 = \bar{Y}_{\text{weekday}}, \qquad \hat{\beta}_1 = \bar{X}_{\text{weekend}} - \bar{Y}_{\text{weekday}}.$$

Moreover, the residual sum of squares equals the pooled sum of squares,

$$\text{RSS} = (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2,$$

and, under the classical homoscedastic OLS assumptions, the t-statistic for testing $H_0 : \beta_1 = 0$ is *identical* to the pooled two-sample t-statistic (5), with matching degrees of freedom $n - 2 = n_1 + n_2 - 2$.

Appendix C gives the full proof. The intuition behind the theorem: the intercept $\hat{\beta}_0$ absorbs the baseline (weekday) mean, and the slope $\hat{\beta}_1$ measures exactly the "jump" when switching groups.

The coefficient $\hat{\beta}_1$'s t-statistic is *identical* to T from Theorem 2.1, testing $H_0 : \beta_1 = 0$ in the regression and testing $H_0 : \mu_1 = \mu_2$ in the two-sample setting are the same hypothesis test, with the same observed value t_{obs}, the same degrees of freedom $n_1 + n_2 - 2$, and therefore the same attained significance level

$$p = 2P(T_{n_1+n_2-2} > |t_{\text{obs}}|)$$

4 The Heteroscedastic Case: When Homogeneity Fails

Real spending data rarely cooperates with the homogeneity assumption: that weekend and weekday spending share a common variance. When $\sigma_1^2 \neq \sigma_2^2$, Theorem 2.1's derivation breaks down, and a different set of tools is required.

4.1 The Behrens–Fisher Problem

Without a shared variance, S_1^2 and S_2^2 estimate two *different* quantities; they cannot legitimately be pooled into one chi-square-distributed sum. Concretely, Step 1 of Theorem 2.1's proof still holds with σ_1^2, σ_2^2 in place of a common σ^2:

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

But Step 2 breaks: $(n_1 - 1)S_1^2/\sigma_1^2 \sim \chi^2_{n_1-1}$ and $(n_2 - 1)S_2^2/\sigma_2^2 \sim \chi^2_{n_2-1}$ are chi-square only after dividing by their *own* variances, and the natural plug-in quantity

$$S_1^2/n_1 + S_2^2/n_2 = \frac{\sigma_1^2}{n_1(n_1-1)}\chi^2_{n_1-1} + \frac{\sigma_2^2}{n_2(n_2-1)}\chi^2_{n_2-1}$$

is a sum of *differently scaled* chi-squares, which is not itself proportional to any χ^2 distribution unless $\sigma_1^2 = \sigma_2^2$. Consequently the natural test statistic

$$T' = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

has no exact, closed-form null distribution[2]for any finite sample size: this is the classical *Behrens–Fisher problem*, and no exact solution exists in general.

4.2 Welch–Satterthwaite Approximation

Welch's fix approximates the null distribution of T' by a t-distribution with a data-dependent, generally non-integer, *effective* degrees of freedom, derived as follows. Approximate the denominator's squared value, $U = S_1^2/n_1 + S_2^2/n_2$, by a scaled chi-square cW', $W' \sim \chi_v^2$, choosing c, v to match the first two moments of U. Since $\mathrm{E}[S_i^2] = \sigma_i^2$ and $\mathrm{Var}(S_i^2) = \dfrac{2\sigma_i^4}{n_i - 1}$ for normal samples,

$$\mathrm{E}[U] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \mathrm{Var}(U) = \frac{2}{n_1 - 1}\left(\frac{\sigma_1^2}{n_1}\right)^2 + \frac{2}{n_2 - 1}\left(\frac{\sigma_2^2}{n_2}\right)^2.$$

Matching moments of cW' ($\mathrm{E} = cv$, $\mathrm{Var} = 2c^2 v$) to $\mathrm{E}[U], \mathrm{Var}(U)$ and solving for v gives

$$v = \frac{2(\mathrm{E}[U])^2}{\mathrm{Var}(U)} = \frac{\left(S_1^2/n_1 + S_2^2/n_2\right)^2}{\dfrac{(S_1^2/n_1)^2}{n_1 - 1} + \dfrac{(S_2^2/n_2)^2}{n_2 - 1}} \tag{8}$$

(plugging in sample variances for the unknown σ_i^2), and the approximation $T' \overset{\cdot}{\sim} t_v$ is Welch's test.

The general formula (8) has two moving parts at once — unequal variances *and* unequal sample sizes — which makes it hard to see which of the two actually drives Welch's correction away from the pooled result. Setting $n_1 = n_2 = n$ removes the second moving part and isolates the first:

Remark 4.1 (Balanced designs: pooled and Welch coincide)
When $n_1 = n_2 = n$, the pooled and Welch standard errors are algebraically identical:

$$S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right) = \frac{S_1^2 + S_2^2}{2} \cdot \frac{2}{n} = \frac{S_1^2}{n} + \frac{S_2^2}{n} = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2},$$

so the two t-statistics are identical; only the degrees of freedom can differ. Substituting $n_1 = n_2 = n$ into (8) gives

$$v/(n_1 + n_2 - 2) = \frac{1}{2}(S_1^2 + S_2^2)^2/(S_1^4 + S_2^4),$$

which depends only on the ratio S_1^2/S_2^2, not on n. This does *not* mean the two tests give the same p-value. Both tests still use the same formula from $p = 2P(T > |t_{\text{obs}}|)$, but evaluated in different reference distributions: the pooled test uses $T \sim t_{n_1+n_2-2}$ exactly, while Welch uses $T \sim t_v$. Since

$$v \le n_1 + n_2 - 2, \qquad \text{(with equality only when } S_1^2 = S_2^2\text{)},$$

[2]The *null distribution* of a test statistic is its distribution under H_0, and it is what a p-value is computed from. For T in Theorem 2.1, the null distribution is $t_{n_1+n_2-2}$, a named, tabulated distribution, so $p = 2P(T_{n_1+n_2-2} > |t_{\text{obs}}|)$ is computed exactly. For T', no such named distribution exists; Welch's approximation is to find a t-distribution that closely matches T''s true behavior, and substitute that in place of the (unknown) exact null distribution.

Welch's reference distribution has *fewer* degrees of freedom, and therefore heavier tails[3]. Comparing the same t_{obs} against a heavier-tailed distribution assigns it *more* tail probability, giving a slightly larger p-value whenever $S_1^2 \neq S_2^2$ — exactly what Section 5 finds numerically ($p = 0.7077$ pooled vs. 0.7079 Welch, for identical $t = 0.3777$).

5 Empirical Validation on Real Data

We apply this model to the study of weekend versus weekday shopper spending, using the UCI "Online Retail" dataset (Chen, 2015), 541,909 real, timestamped line items from a UK online retailer, Dec 2010–Dec 2011.

5.1 A Realistic Sample Against the Full Population

We compare a realistic $n_1 = n_2 = 500$ sample against the full population it was drawn from, at each stage of the same trimming procedure. Trimming here refers to the outlier-removal step described earlier: writing Q_1, Q_3 for the first and third quartiles of the combined sample, orders exceeding

$$Q_3 + 1.5(Q_3 - Q_1)$$

are excluded, with this same cutoff applied to both groups.

Stage	n_1/n_2	mean gap	p (pooled)	p (Welch)
Raw sample	500/500	\$89.78	0.01073	0.01077
Trimmed sample	474/444	\$13.58	0.3067	0.3072
Full population (untrimmed)	2,204/17,755	\$185.06	4.08×10^{-6}	3.26×10^{-24}
Full population (trimmed)	2,117/16,032	\$10.07	**0.0454**	**0.0337**

Remark 5.1 (A non-significant sample result is not evidence of no difference)
A researcher working only with the trimmed sample (474/444) would compute $p \approx 0.31$ and could reasonably, but incorrectly, conclude "no evidence of a difference, weekend and weekday shoppers are essentially the same." At the full-population scale, the exact same trimming procedure still yields $p < 0.05$ by both tests (pooled 0.0454, Welch 0.0337): a real difference does exist. The correct conclusion from the sample alone should have been "not enough evidence to detect a difference," not "no difference exists".

Remark 5.2 (Pooled/Welch divergence tracks both imbalance and heteroscedasticity)
The rows above trace out Remark 4.1's balanced-design result across a wide range of imbalance. At $n_1 = n_2 = 500$ exactly, the pooled and Welch t-statistics coincide numerically ($t = -2.5561$ both), differing only in degrees of freedom (998 vs. 781.55), exactly as Remark 4.1 predicts. At $n_1 = 474, n_2 = 444$ — close but no longer exactly equal — the coincidence is close but no longer exact ($t = 1.0227$ vs. 1.0218). At the population level, the divergence grows sharply with the imbalance ($n_1 \approx 2{,}200$ against $n_2 \approx 18{,}000$): part of this is because n_1 and n_2 are very different. But the size of the divergence here is also a signal that the assumption underlying the pooled test about **equal variance across weekend and weekday spending may not hold.**

6 Conclusion

The two-sample t-test is not a separate technique from linear regression; it is linear regression with one binary predictor. Variance homogeneity is exactly the classical OLS homoscedasticity

[3] A t-distribution's tails grow heavier — assign more probability to large $|t|$ — as its degrees of freedom decrease.

assumption, and it earns you an exact, finite-sample t distribution (Theorem 2.1). When homogeneity fails, the Behrens–Fisher problem shows the natural test statistic has no exact null distribution, and Welch's approximation fixes this at the cost of a data-dependent, generally non-integer degrees of freedom; in a balanced design the pooled and Welch t-statistics coincide exactly, yet their p-values still diverge whenever the two variances differ (Remark 4.1). Applied to 541,909 real transaction records, a realistic 500-per-group sample illustrates two risks a textbook problem hides by construction: a non-significant sample result can reflect insufficient power rather than a genuine absence of a difference (Remark 5.1), and the gap between the pooled and Welch tests widens both mechanically, as group sizes grow unequal, and substantively, as a signal of real variance heterogeneity in the data (Remark 5.2). Every theorem in this document is proved at the point it is stated, and every empirical claim is checked against real data rather than asserted.

Appendix A Distributional Facts for Normal Samples

Definition A.1 (Chi-square distribution)
Let $Z_1, \dots, Z_k$ be independent and identically distributed standard normal random variables,

$$Z_1, \dots, Z_k \overset{iid}{\sim} N(0,1).$$

The distribution of the random variable

$$Q = \sum_{i=1}^{k} Z_i^2$$

is called the chi-square distribution with k degrees of freedom, denoted χ^2_k.

Lemma A.1 (Additivity of chi-square)
Let $U \sim \chi^2_{k_1}$ and $V \sim \chi^2_{k_2}$ be independent random variables. Then

$$U + V \sim \chi^2_{k_1+k_2}.$$

Proof.

By definition, since U and V are chi-square, there exist i.i.d. standard normal random variables $Z_1, \dots, Z_{k_1}$ and $W_1, \dots, W_{k_2}$ such that

$$U = \sum_{i=1}^{k_1} Z_i^2, \qquad V = \sum_{j=1}^{k_2} W_j^2.$$

Independence of U and V allows all $k_1 + k_2$ variables $Z_1, \dots, Z_{k_1}, W_1, \dots, W_{k_2}$ to be taken mutually independent. Setting $(T_1, \dots, T_{k_1+k_2}) := (Z_1, \dots, Z_{k_1}, W_1, \dots, W_{k_2})$, these are i.i.d. $N(0,1)$, and

$$U + V = \sum_{i=1}^{k_1} Z_i^2 + \sum_{j=1}^{k_2} W_j^2 = \sum_{\ell=1}^{k_1+k_2} T_\ell^2.$$

By definition of the chi-square distribution, $U + V \sim \chi^2_{k_1+k_2}$. □

Theorem A.1 (Distribution of the sample variance, normal case)
Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, and define

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Then $\bar{X}$ and S^2 are independent, and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Proof.

Define $Z_i = (X_i - \mu)/\sigma$ for $i = 1, \dots, n$, so that

$$Z_1, \dots, Z_n \overset{iid}{\sim} N(0,1), \qquad \bar{Z} := \frac{1}{n}\sum_{i=1}^{n} Z_i = \frac{\bar{X}-\mu}{\sigma}.$$

Let O be an $n \times n$ orthogonal matrix ($OO^\top = O^\top O = I_n$) whose first row is

$$o_1 = \left(\frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}}\right),$$

completed arbitrarily to a full orthonormal basis of $\mathbb{R}^n$ (possible via Gram–Schmidt procedure) and set

$$V = OZ, \qquad Z = (Z_1, \dots, Z_n)^\top, \qquad V = (V_1, \dots, V_n)^\top.$$

Since $Z \sim N(0, I_n)$ and O is orthogonal,

$$V = OZ \sim N\Big(O \cdot 0,\ OI_nO^\top\Big) = N(0,\ OO^\top) = N(0, I_n),$$

so $V_1, \dots, V_n$ are themselves i.i.d. $N(0,1)$, and by construction of the first row of O,

$$V_1 = o_1^\top Z = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i = \sqrt{n}\,\bar{Z}.$$

Orthogonality also preserves the sum of squares,

$$\sum_{i=1}^{n} V_i^2 = V^\top V = Z^\top O^\top O Z = Z^\top Z = \sum_{i=1}^{n} Z_i^2.$$

Expanding the centered sum of squares,

$$\sum_{i=1}^{n}(Z_i - \bar{Z})^2 = \sum_{i=1}^{n} Z_i^2 - 2\bar{Z}\sum_{i=1}^{n} Z_i + n\bar{Z}^2 = \sum_{i=1}^{n} Z_i^2 - 2n\bar{Z}^2 + n\bar{Z}^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2,$$

and substituting $\sum_i Z_i^2 = \sum_i V_i^2$ and $n\bar{Z}^2 = V_1^2$ from above,

$$\sum_{i=1}^{n}(Z_i - \bar{Z})^2 = \sum_{i=1}^{n} V_i^2 - V_1^2 = \sum_{i=2}^{n} V_i^2.$$

Since $V_1, \dots, V_n$ are i.i.d., V_1 is independent of $(V_2, \dots, V_n)$, hence of any function of $(V_2, \dots, V_n)$ alone, in particular of $\sum_{i=2}^{n} V_i^2$. As $\bar{Z}$ (and hence $\bar{X} = \sigma\bar{Z} + \mu$) is a function of V_1 alone, while $\sum_{i=2}^{n} V_i^2 = \sum_i (Z_i - \bar{Z})^2$ (and hence S^2) is a function of $(V_2, \dots, V_n)$ alone, it follows that the *sample mean* $\bar{X}$ and the *sample variance* S^2 are *independent.*

Finally, $V_2, \dots, V_n$ are $n-1$ i.i.d. standard normal random variables, and

$$\sum_{i=2}^{n} V_i^2 = \sum_{i=1}^{n}(Z_i - \bar{Z})^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{(n-1)S^2}{\sigma^2}.$$

By definition of the chi-square distribution, $\sum_{i=2}^{n} V_i^2 \sim \chi^2_{n-1}$, and therefore

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

□

Appendix B The Student t-Distribution as a Normal-over-Chi-Square Ratio

Definition B.1 (Student's t-distribution)
A random variable T has the Student t-distribution with $k > 0$ degrees of freedom, written $T \sim t_k$, if its probability density function is

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}}, \qquad t \in \mathbb{R},$$

where Γ is the Gamma function.

Theorem B.1 (Normal-over-chi-square ratio is Student t)
Let $Z \sim N(0,1)$ and $W \sim \chi^2_k$ be independent. Then

$$T := \frac{Z}{\sqrt{W/k}} \sim t_k.$$

Proof.

The density of Z is $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, and the density of W, a standard fact for the chi-square distribution (a special case of the Gamma family), is

$$f_W(w) = \frac{1}{2^{k/2}\Gamma(k/2)} w^{k/2-1} e^{-w/2}, \qquad w > 0.$$

Since Z and W are independent, their joint density is $f_{Z,W}(z,w) = \varphi(z) f_W(w)$.

Change variables from (Z,W) to (T,W) via $Z = T\sqrt{W/k}$, holding W fixed; for fixed w this map is linear and invertible in t, with Jacobian $\partial z/\partial t = \sqrt{w/k}$. The joint density of (T,W) is therefore

$$f_{T,W}(t,w) = \varphi\left(t\sqrt{w/k}\right) f_W(w) \sqrt{w/k}.$$

Integrating out w gives the marginal density of T:

$$f_T(t) = \int_0^\infty \varphi\left(t\sqrt{w/k}\right) f_W(w)\sqrt{w/k}\,dw = \frac{1}{\sqrt{2\pi k}\,2^{k/2}\Gamma(k/2)} \int_0^\infty w^{(k-1)/2} \exp\left(-\frac{w}{2}\left(1 + \frac{t^2}{k}\right)\right) dw.$$

Writing $a = \frac{1}{2}\left(1 + \frac{t^2}{k}\right)$ and using the Gamma integral $\int_0^\infty w^{s-1} e^{-aw}\,dw = \Gamma(s)/a^s$ with $s = \frac{k+1}{2}$,

$$\int_0^\infty w^{(k-1)/2} e^{-aw}\,dw = \frac{\Gamma\left(\frac{k+1}{2}\right)}{a^{(k+1)/2}}.$$

Substituting back,

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{2\pi k}\,2^{k/2}\Gamma(k/2)} \cdot \left(\frac{2}{1 + t^2/k}\right)^{(k+1)/2} = \frac{\Gamma\left(\frac{k+1}{2}\right) 2^{1/2}}{\sqrt{2\pi k}\,\Gamma(k/2)} \left(1 + \frac{t^2}{k}\right)^{-(k+1)/2}.$$

Since $2^{1/2}/\sqrt{2\pi k} = 1/\sqrt{k\pi}$, this simplifies to

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}},$$

which is exactly the density of t_k, proving $T \sim t_k$. □

Remark B.1 (Where the independence hypothesis comes from)
Theorem B.1 requires Z and W to be independent; this is not a technicality to wave away, since the entire derivation above rests on it (the change of variables treats W as fixed while integrating over Z, which is only valid if W's distribution does not depend on Z). Theorem A.1 in Appendix A is precisely where this independence was established: it showed not only that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, but also, as part of the same argument, that $\bar{X}$ and S^2 are independent. This is exactly the hypothesis Theorem B.1 needs to apply $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ and $W = (n-1)S^2/\sigma^2$ as an independent normal/chi-square pair, and is what licenses Step 3 of Theorem 2.1's proof, where the two-sample version of this same argument is used.

Appendix C General Least Squares Theory and the Regression/Two-Sample Equivalence

C.1 The Normal Equations

The least-squares estimator minimizes

$$Q(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} + \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}.$$

Setting $\nabla_{\boldsymbol{\beta}} Q = -2\mathbf{X}^\top\mathbf{y} + 2\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} = 0$ gives the normal equations $\mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y}$, and since $2\mathbf{X}^\top\mathbf{X}$ is positive semi-definite (positive definite when $\mathbf{X}^\top\mathbf{X}$ is invertible), the unique global minimizer is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y},$$

which is (7).

C.2 Proof of Theorem 3.1

Proof.

Let $n_1 = \sum_i D_i$ (number of weekend observations) and $n_2 = n - n_1$ (weekday observations). Since $D_i \in \{0, 1\}$, $D_i^2 = D_i$, so

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{y} = \begin{pmatrix} \sum_i Y_i \\ \sum_i D_i Y_i \end{pmatrix} = \begin{pmatrix} n_1\bar{X} + n_2\bar{Y} \\ n_1\bar{X} \end{pmatrix},$$

writing $\bar{X}$ for the mean of the weekend ($D_i = 1$) observations and $\bar{Y}$ for the mean of the weekday ($D_i = 0$) observations. The normal equations $\mathbf{X}^\top\mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{y}$ read

$$\begin{aligned} n\hat{\beta}_0 + n_1\hat{\beta}_1 &= n_1\bar{X} + n_2\bar{Y}, \\ n_1\hat{\beta}_0 + n_1\hat{\beta}_1 &= n_1\bar{X}. \end{aligned}$$

Dividing the second equation by n_1 (assuming $n_1 > 0$) gives $\hat{\beta}_0 + \hat{\beta}_1 = \bar{X}$, i.e. $\hat{\beta}_0 = \bar{X} - \hat{\beta}_1$. Substituting into the first equation:

$$n(\bar{X} - \hat{\beta}_1) + n_1\hat{\beta}_1 = n_1\bar{X} + n_2\bar{Y} \implies n\bar{X} - (n - n_1)\hat{\beta}_1 = n_1\bar{X} + n_2\bar{Y} \implies n\bar{X} - n_2\hat{\beta}_1 = n_1\bar{X} + n_2\bar{Y}.$$

Since $n = n_1 + n_2$, $n\bar{X} - n_1\bar{X} = n_2\bar{X}$, so $n_2\bar{X} - n_2\hat{\beta}_1 = n_2\bar{Y}$, giving $\hat{\beta}_1 = \bar{X} - \bar{Y}$ and therefore $\hat{\beta}_0 = \bar{X} - \hat{\beta}_1 = \bar{Y}$. This proves the first claim.

Residual sum of squares. The fitted value for a weekend observation ($D_i = 1$) is $\hat{\beta}_0 + \hat{\beta}_1 = \bar{Y} + (\bar{X} - \bar{Y}) = \bar{X}$; for a weekday observation ($D_i = 0$) it is $\hat{\beta}_0 = \bar{Y}$. Hence

$$\text{RSS} = \sum_{i:D_i=1} (Y_i - \bar{X})^2 + \sum_{i:D_i=0} (Y_i - \bar{Y})^2 = (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2,$$

exactly the numerator of the pooled variance S_p^2 in (5). Since the model has 2 parameters, the residual degrees of freedom are $n - 2 = n_1 + n_2 - 2$, so the OLS variance estimator is $\hat{\sigma}^2 = \text{RSS}/(n-2) = S_p^2$ — *identical* to the pooled variance estimator.

Standard error and t-statistic. By direct multiplication, $\det(\mathbf{X}^\top\mathbf{X}) = nn_1 - n_1^2 = n_1(n - n_1) = n_1n_2$, so

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{n_1n_2}\begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix}, \tag{9}$$

whose $(2, 2)$ entry is $n/(n_1 n_2) = 1/n_1 + 1/n_2$. Under homoscedasticity, $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \cdot (1/n_1 + 1/n_2)$, estimated as $\widehat{\mathrm{Var}}(\hat{\beta}_1) = S_p^2(1/n_1 + 1/n_2)$, so

$$\widehat{\mathrm{SE}}(\hat{\beta}_1) = S_p\sqrt{1/n_1 + 1/n_2}, \qquad t = \frac{\hat{\beta}_1 - 0}{\widehat{\mathrm{SE}}(\hat{\beta}_1)} = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{1/n_1 + 1/n_2}},$$

which is exactly (5), with the same degrees of freedom $n - 2 = n_1 + n_2 - 2$. This completes the proof. $\square$